Application of Remote Sensing Technologies to Map the Structural Geology of Central Region of Kenya

Mercy W. Mwaniki, Moeller S. Matthias, and Gerhard Schellmann

***Abstract*—Advancements of digital image processes (DIP) and availability of multispectral and hyperspectral remote sensing data have greatly benefited mineral investigation, structure geology mapping, fault pattern, and landslide studies: site-specific landslide assessment and landslide quantification. The main objective of this research was to map the geology of the central region of Kenya using remote-sensing techniques to aid rainfall-induced landslide quantification. The study area is prone to landslides geological hazards and, therefore, it was necessary to investigate geological characteristics in terms of structural pattern, faults, and river channels in a highly rugged mountainous terrain. The methodology included application of PCA, band rationing, intensity hue saturation (IHS) transformation, ICA, false color composites (FCC), filtering applications, and thresholding, and performing knowledge-based classification on Landsat ETM+ imagery. PCA factor loading facilitated the choice of bands with the most geological information for band rationing and FCC combination. Band ratios (3/2, 5/1, 5/4, and 7/3) had enhanced contrast on geological features and were the input variables in a knowledge-based geological classification. This was compared to a knowledge-based classification using PCs 2, 5, and IC1, where the band ratio classification performed better at representing geology and matched FCC [IC1, PC5, saturation band of IHS (5,7,3)]. Fault and lineament extraction was achieved by filtering and thresholding of pan-band8 and ratio 5/1 and overlaid on the geology map. However, the best visualization of lineaments and geology was in the FCC [IC1, PC5, saturation band of IHS (5,7,3)], where volcanic extrusions, igneous, sedimentary rocks (eolian and organic), and fluvial deposits were well discriminated.**

***Index Terms*—Digital image processing (DIP), false color composites (FCC), independent component (IC), intensity hue saturation (IHS), principal component analysis (PCA).**

I. Introduction

DIGITAL image processing (DIP) in remote sensing has greatly boosted geology and mineralization studies in lithological discrimination of rocks, delineation of structural, geological features, and hydrothermal-altered rock deposits. Availability of satellites such as ASTER, hypersion–hyperspectral imager, and Landsat providing data in the visible, near-infrared, and shortwave-infrared regions has proved very useful in geological and mineral exploration studies in lithological discrimination of rocks and delineation of geological structural features. Each multispectral band records unique energy interaction with a surface and thus remote-sensing interpretations are made based on the spectral signatures, color, and texture to distinguish the different minerals and elements comprising rocks and soils [1].

Geological features are enhanced spectrally (through techniques such as: linear stretching, principal component analysis (PCA), decorrelation stretch, RGB color combinations, band rationing, density slicing) and spatially (through image fusion and filtering), thereby improving their tones, hues, image texture, fracture patterns, lineaments, and trends which aid geological interpretation and classification [2]. Image enhancement methods produce new images with detailed information from the highly correlated bands. According to Abdeen and Abdelghaffar [3], bands containing most geological information are highly correlated, as they occupy only a small part of the spectral range.

The main aim of carrying out this study was to utilize remote-sensing techniques to map the geology of the central region of Kenya and to develop remote-sensing methods, which can be used to update the existing geology maps especially in landslide-prone areas. The study area has highly rugged terrain with deep incised river channels, as it contains three most important Kenya's water towers and hence the rivers form dendritic drainage pattern as they flow to the lower regions. Geology map exist at small scale of about 1:250 000 covering the whole country and is insufficient since the area experiences landslides at least every once in 3 years. An attempt to utilize remote sensing method to map geology is by [4] only covering Nairobi and investigating the swelling of soils. Landslide studies in the study area in [5]–[8] have described and documented the landslide-causing factors in the area; among them are high-absorbent clays, rainfall triggers, and human activities. Therefore, there is need to improve the soil and geology maps for proper landslide disaster management.

II. Use of Remote Sensing DIP and Landsat Imagery in Geology Applications

The availability of higher spectral resolution satellite imagery, covering VNIR and SWIR spectral regions such as Landsat and ASTER, has boosted geological mapping at small scales and cheaply compared to conventional geological mapping. Further, the improvement of multispectral satellites with a higher spatial resolution panchromatic band or higher spatial multispectral resolution (such as worldview-2

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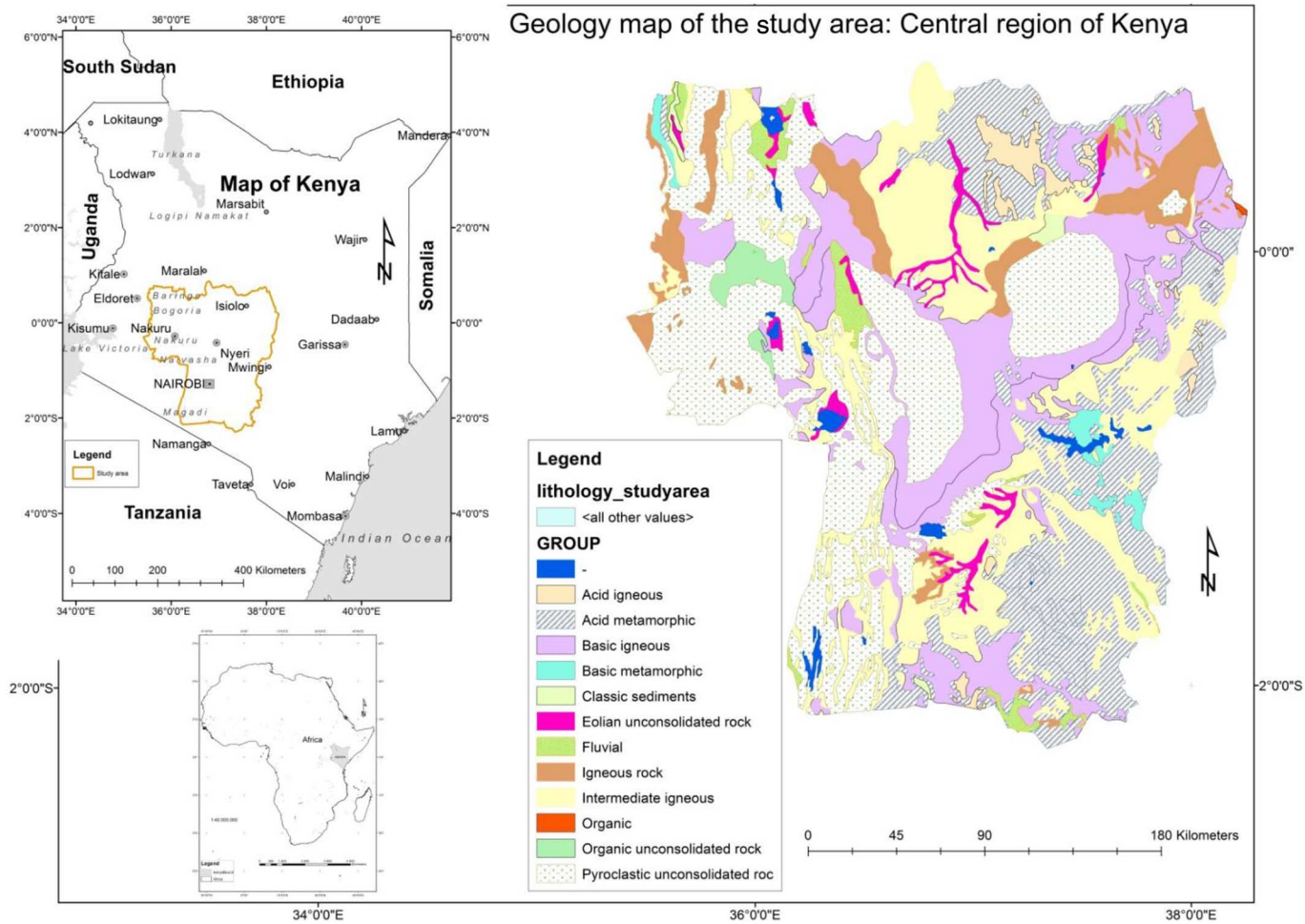

Fig. 1. Geology map of the study area.

satellite) increases the accuracy of the lineaments extracted. For example, Hung *et al.* [9] compared the lineaments obtained from 15-m spatial resolution [ASTER bands (1,2,3) and Landsat ETM+ fused bands (3,4,5)] and 30-m spatial resolution (ASTER SWIR bands and Landsat ETM+ bands 1–5, and 7) and found that the former had nearly twice the lineaments obtained from the latter. Landsat was also compared to SPOT satellite in a study by Newton and Boyle [10], where it was found that Landsat TM was more superior than SPOT in lithological applications although, SPOT has higher spatial resolution than Landsat, and vice versa for spectral resolution.

DIP enhancements suiting geological applications have exploited the strength of more spectral information using methods such as spectral signature analysis (SSA), PCA, ICA, band rationing, FCC, and image-fusion methods to discriminate and extract geological information. SSA is the visual analysis of multispectral data in a reflectance spectrum, so that a single pixel is seen through many bands. Abdeen and Hassan [11] applied SSA followed by PCA to select the most appropriate band combination for discriminating sands and gravels. PCA works by decorrelating bands, reducing noise, and separating geologic features along the new principal components thus aiding classification of rocks. Application of FCC applying PCA or band rationing has also proved effective in lithological and structural mapping utilizing, and maximizing on color differences arising from minerals comprising the rocks. For example, Abdeen and Abdelghaffar [3] implemented FCC of PCs (1,2,3) and band ratios (5/7,5/1, $5/4 * 3/4$) and (5/7,7/5, $5/4 * 3/4$) using Landsat ETM+. A photogeological map was produced by density-slicing the gray-scale values of the four band ratios used (5/7, 7/5, 5/1, $5/4 * 3/4$).

False color composite (FCC) is one of the best ways to visually interpret a multispectral image [12] and it can utilize individual bands or band ratios. Use of color composites requires the selection of three bands, which are individually informative and collectively least-correlated [13], [14]. Thus, methods such as PCA, optimum index factor [15], and visual inspection of feature space images are commonly used to determine band combinations with less correlation. Examples of such band combinations include: (5,3,1) used by Crósta and Moore [16], (5,4,1) used by Newton and Boyle [10], (5,4,3) and (7,4,1) used by Ali *et al.* [13] and (3,2,1) used by Wahid and Ahmed [17] in marine geology. Color composites can also involve band ratios, e.g., Kavak [18] used band ratio (5/7, 5/4, 4/1) in FCC to emphasize the lithologic differences in an arid area.

Band rationing works to reduce effects of relief and shadowing while extracting and emphasizing the differences in

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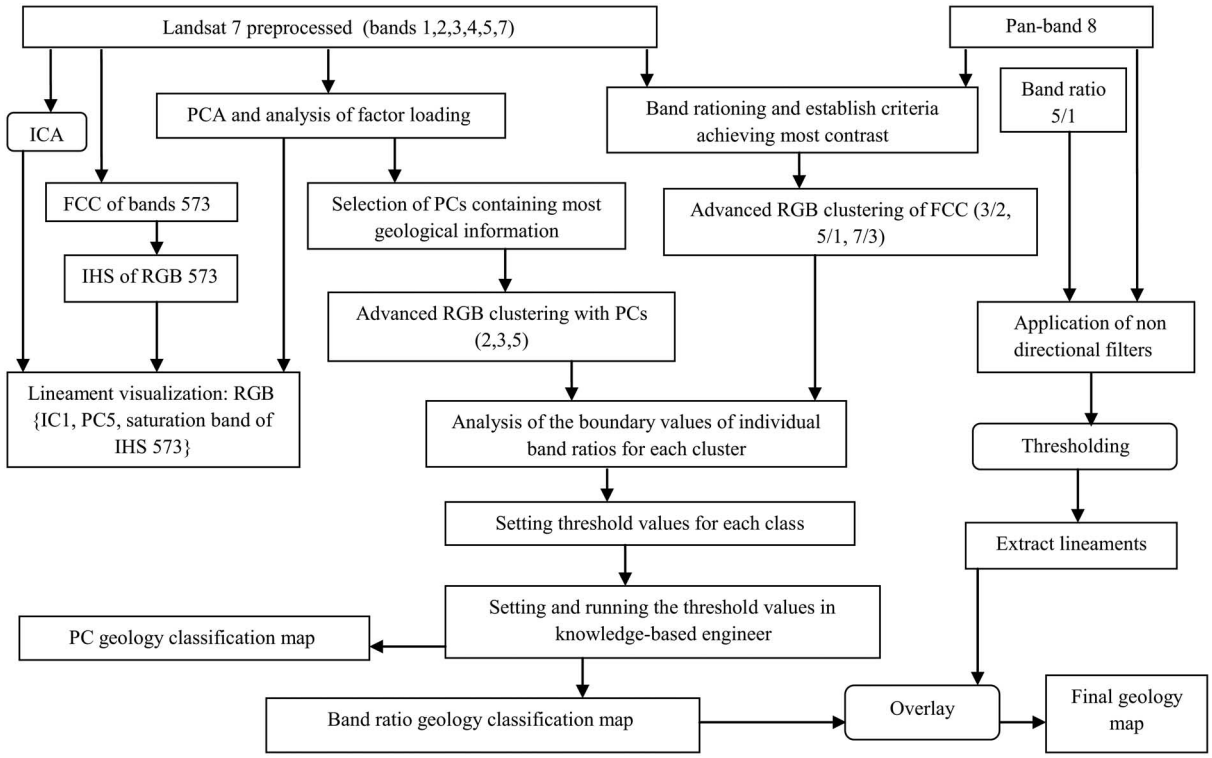

Fig. 2. Summary of methodology flowchart.

TABLE I
PC FACTOR LOADING COMPUTED FROM COVARIANCE_VARIANCE MATRIX

	PC1	PC2	PC3	PC4	PC5	PC7
eigvec.1	0.3504299	0.14994187	0.3435107	−0.066525	0.4175276	−0.7469684
eigvec.2	0.3118657	0.17774095	0.41251398	−0.095809	0.5037826	0.66181866
eigvec.3	0.4063747	−0.1377272	0.54636037	0.3519655	−0.6264414	0.03275176
eigvec.4	0.2829485	0.82871519	−0.2396491	−0.241326	−0.3422539	0.01906948
eigvec.5	0.5846698	−0.1363933	−0.5776030	0.5147799	0.1971126	0.0456196
eigvec.7	0.4392034	−0.4707280	−0.1492179	−0.734356	−0.1531462	0.0227320
Eigenvalues	14486.5364	286.7156	176.9544	25.5547	11.3763	3.3523
% Var	96.64	1.91	1.18	0.17	0.08	0.02

spectral reflectance of materials [19]. Particular Landsat 7 band ratios are known for rock discrimination based on the mineral composition. Examples are Kaufmann ratio (7/4, 4/3, 5/7), Chica–Olma ratio (5/7, 5/4, 3/1), and Abrams ratio (5/7, 3/1, 4/5) [20]. Further, the multiplication of band ratios maximizes rock discrimination, since the individual bands ratios are sensitive to specific chemical and mineral components of the rock [13]. An example of multiplicative band ratio is $5/4 * 3/4$, which is used in the Sultan's color composite ratio (5/7, 5/1, $5/4 * 3/4$) in [21] to map metavolcanic rocks.

Utilization of band ratios have been emphasized by several geological researchers, e.g., Sabins [22] used ratios 3/1, 5/1, and 5/7 to discriminate iron oxides, magnetite content, and hydroxyl bearing (clay minerals) rocks, respectively; whereas, Abdeen and Abdelghaffar [3] used ratio 7/5 and 5/4 to discriminate granitoid felsic rocks from ferrous minerals. This is explained by Campbell [23] in the usefulness of each band, where band 1 suited for water investigation; bands 2 and 4 are high reflective zones for vegetation and, therefore, suited for vegetation analysis; band 3 is helpful for discriminating soil from vegetation due to the high-absorbency effect of vegetation; and bands 5 and 7 are best suited for rock and soil studies, since soil has high absorption in band 7 and high reflectance in band 5. Studies by Crippen [24] and Kenea [25] further used band 1 to provide information on ferric and ferrous iron, band 4 to provide information on iron oxides and hydroxides, and band 7 to provide information on hydroxide-bearing minerals, clay, and layered silicates.

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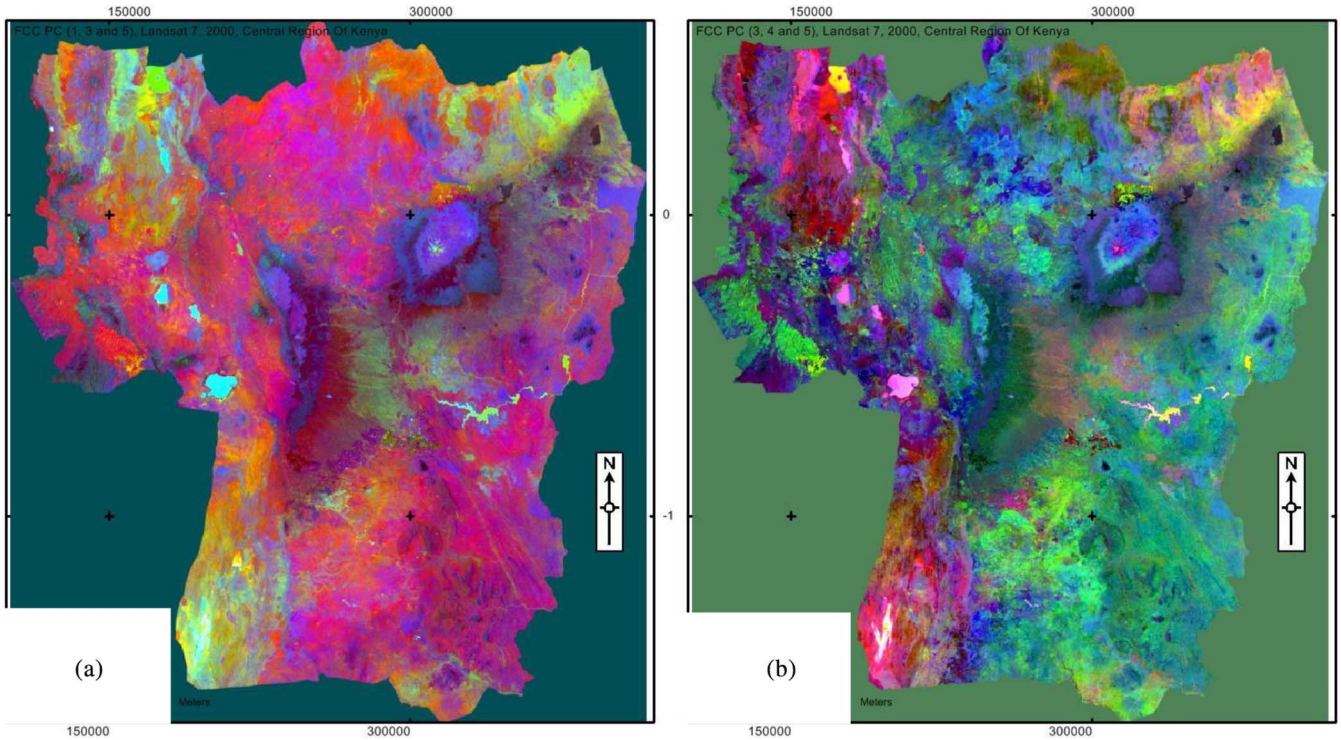

Fig. 3. (a) FCC PC (1,3,5). (b) FCC PC (3,4,5).

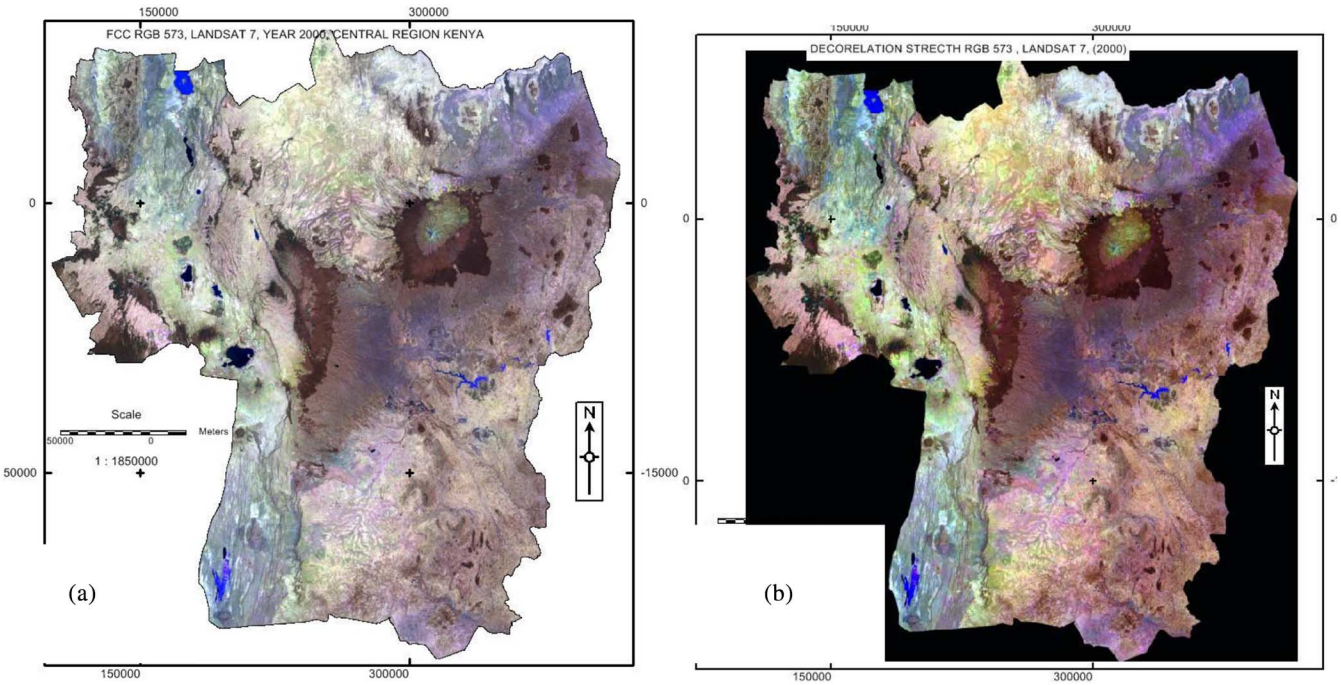

Fig. 4. (a) FCC with bands (5,7,3). (b) FCC (5,7,3) after decorrelation stretch.

Landsat TM data was found by Harris *et al.* [26] to provide useful information with regard to compositional layering, structural patterns, and vegetation mapping. The researcher produced geological and mineral exploration maps using variety of remote-sensed data and applying maximum likelihood and neural network methods of classification. Rawashdeh *et al.* [27] determined igneous rocks from Landsat TM by the use of color composite of PC (1,2,3), the ratio images (3/1, 4/3, and 5/7), and the IHS (5, 3, 1).

Automatic lineament extraction softwares such as PCI GeoAnalyst or Geomatica have further aided lithological mapping. Qari *et al.* [28] extracted the structural information from Landsat ETM+ band 8 using PCI GeoAnalyst software by applying edge detection and directional filtering followed by

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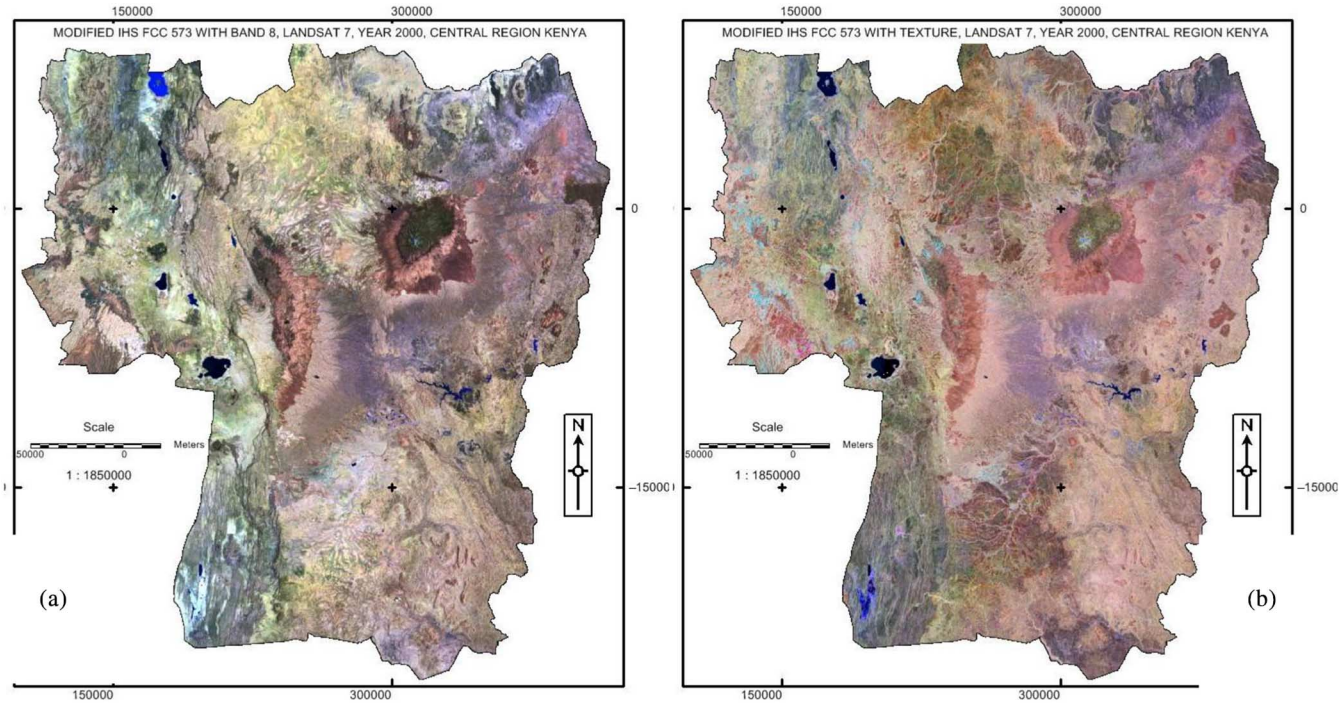

Fig. 5. (a) Pansharpened band 8, FCC 573 in IHS transformation. (b) Pansharpened band 8 edges, FCC 573 in IHS transformation.

overlaying with ASTER band ratios 6/8, 4/8, and 11/14 in RGB to create a geological map. Kocal *et al.* [29] extracted lineaments using line module of PCI Geomatica from band 8 but defined the direction of the lineaments manually. While previous researchers have developed means of mapping geology in arid conditions, applying FCCs of band ratios, and PC bands images, this research aims to establish band ratios for mapping the geology in the central region of Kenya, which has highland to savanna climatic conditions.

III. Methodology

A. Study Area

The study area is the central region of Kenya and ranges from longitude $35°34'00''$E to $38°15'00''$E and latitudes $0°53'00''$N to $2°10'00''$S (Fig. 1). It has a highly rugged mountainous terrain, with deep incised river valleys and narrow ridges, and altitude varying from 450 to 5100 m above mean sea level. The geology of the study area comprises mostly pyroclastic rocks such as tuff, agglomerates, and ashes, which are associated with volcanic formation of Mt. Kenya and Aberdare ranges [8]. Deep weathering of rocks is attributed to soil formation and Ngecu *et al.* [8] noted majorly three types of soils: nitosols, andosols, and cambisol. The climate varies from highland to savanna climatic conditions with forest, agriculture, and settlement being the most prevalent land covers land use. Landslides triggered by rainfall are also a major threat on the south-eastern slopes of Aberdare mountain ranges as reported in [30] and [31].

B. Data Description and Processing Methods

Landsat ETM+ scenes p168r060, p168r061, and p169r060 free of cloud cover for the year 2000 were downloaded from USGS Website page and preprocessed to reduce the effects of haze before mosaicing. Fig. 2 shows the summary flowchart of the methodology following preprocessing where Landsat ETM+ bands were investigated using PCA factor loading to determine bands suitable for geological investigation (Table I). PCs 1, 2, 3, 4, and 5 were found to contain the most geologic information from bands 7, 5, and 3. PC1 had information from all bands positively correlated making it difficult to differentiate the soil from other covers though it had 96.6% of all information. PC7, on the other hand, had its contribution to only 0.02% of all the information and was, therefore, not considered. PC2 had the most vegetation information from band 4 and information from bands 5 and 7 is negatively correlated to band 4, thus facilitating discrimination of geological information from band 7, which had the second highest information. PC3 had high information from bands 5 and 3, which were negatively correlated, thus facilitating discrimination by soil moisture properties. PC4 had information from band 7 negatively correlated to bands 3 and 5 thereby, enabling separation of geological and soil information. PC5 had highest information from band 3 and least information from band 7, while bands 1 and 2 were positively correlated. This facilitated separation of fresh and turbid water while providing soil moisture information in bands 3, 4, and 5. Based on the factor loading, PCP combination 1, 3, 5 [Fig. 3(a)] had the most geological information although PCP combination 3, 4, 5 [Fig. 3(b)] had better enhanced geological features.

An FCC of bands [5, 7, and 3, Fig. 4(a)] was performed and the result improved further using decorrelation stretch [Fig. 4(b)]. IHS transformation of FCC 5,7,3 [Fig. 6(a)] was then performed and modified IHS image fusion with pan-band 8 performed according to Ehlers *et al.* [32], where the intensity band is replaced with pan-band 8 after histogram matching

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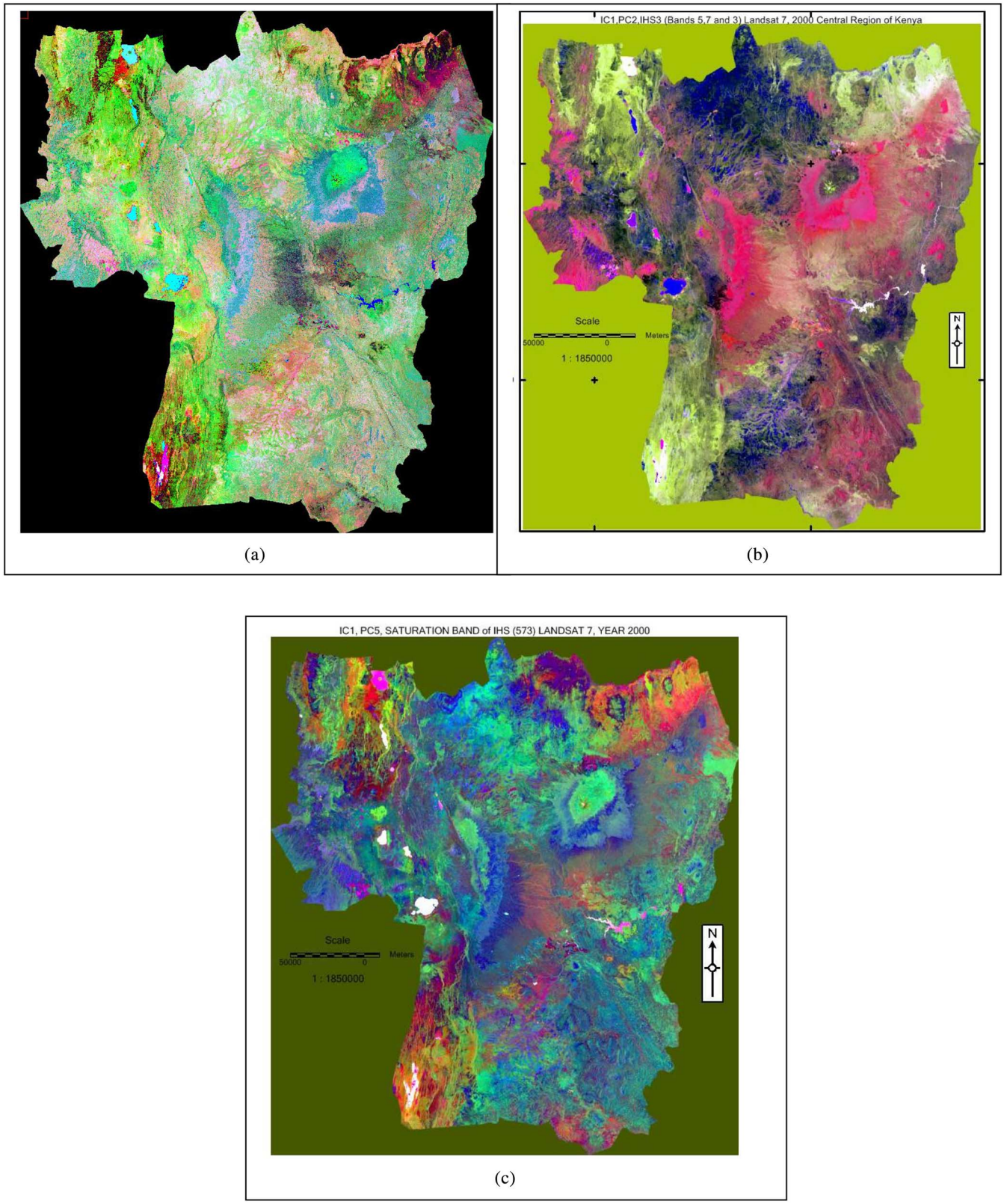

Fig. 6. (a) IHS of FCC 573. (b) IC1, PC2, saturation band of IHS FCC 573. (c) IC1, PC5, saturation band of IHS FCC 573.

the pan-band to the original intensity band [Fig. 5(a)]. Edges were extracted from band 8 through application of nondirectional filters and fused with the FCC (5,7,3) using IHS-modified method [Fig. 5(b)]. Further, the subset image was processed using independent canonical analysis to discriminate geological features better from soil information. An FCC comprising IC1, PC2, and the saturation band of IHS transformation of bands 5,7,3 was layerstacked as in Fig. 6(b) and also with PC5

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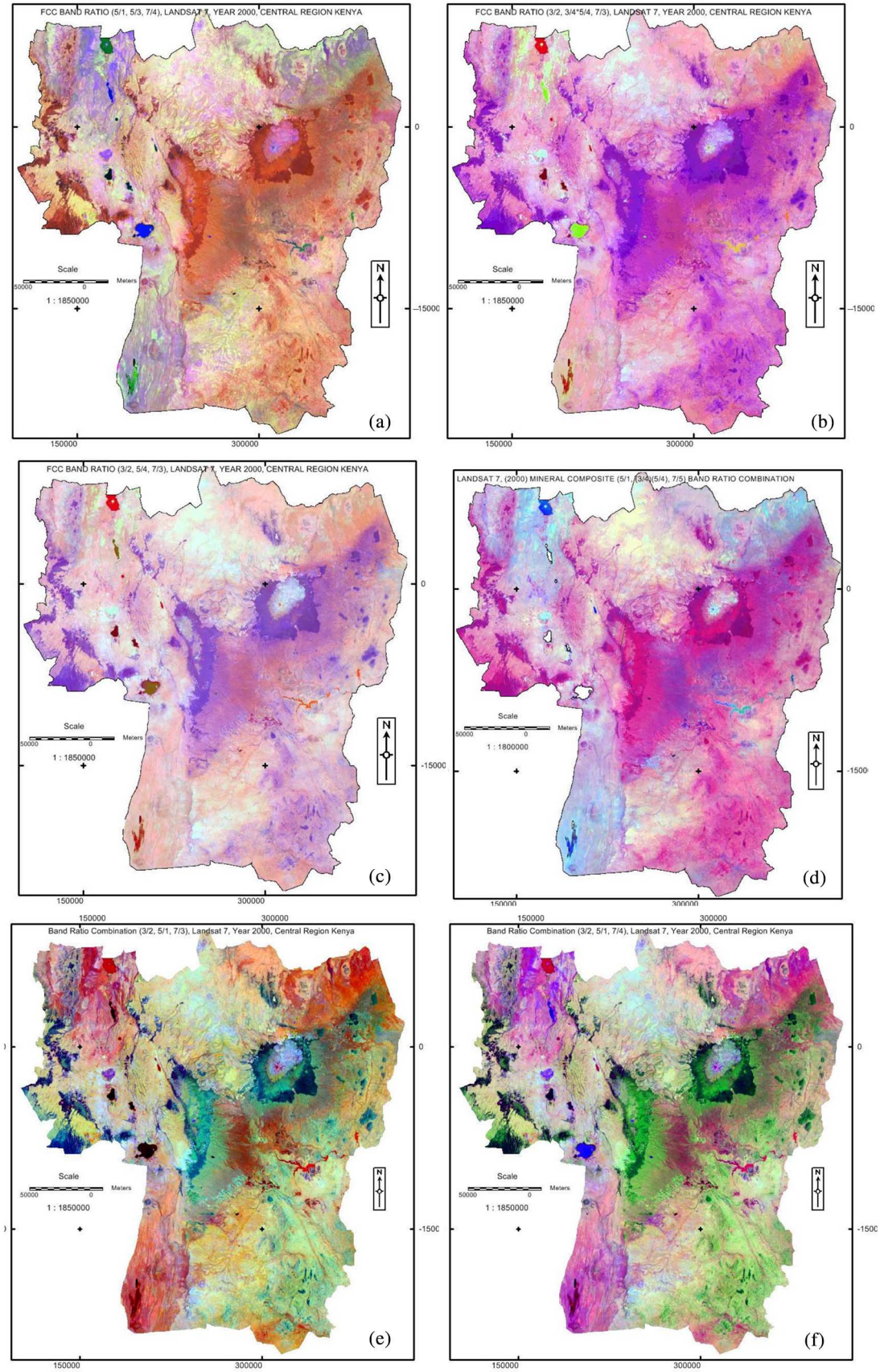

Fig. 7. FCC Band Ratios. (a) (5/1, 5/3, 7/4). (b) (3/2, $3/4 * 5/4$, 7/3). (c) (3/2, 5/4, 7/3). (d) (5/1, $3/4 * 5/4$, 7/5). (e) (3/2, 5/1, 7/3). (f) (3/2, 5/1, 7/4).

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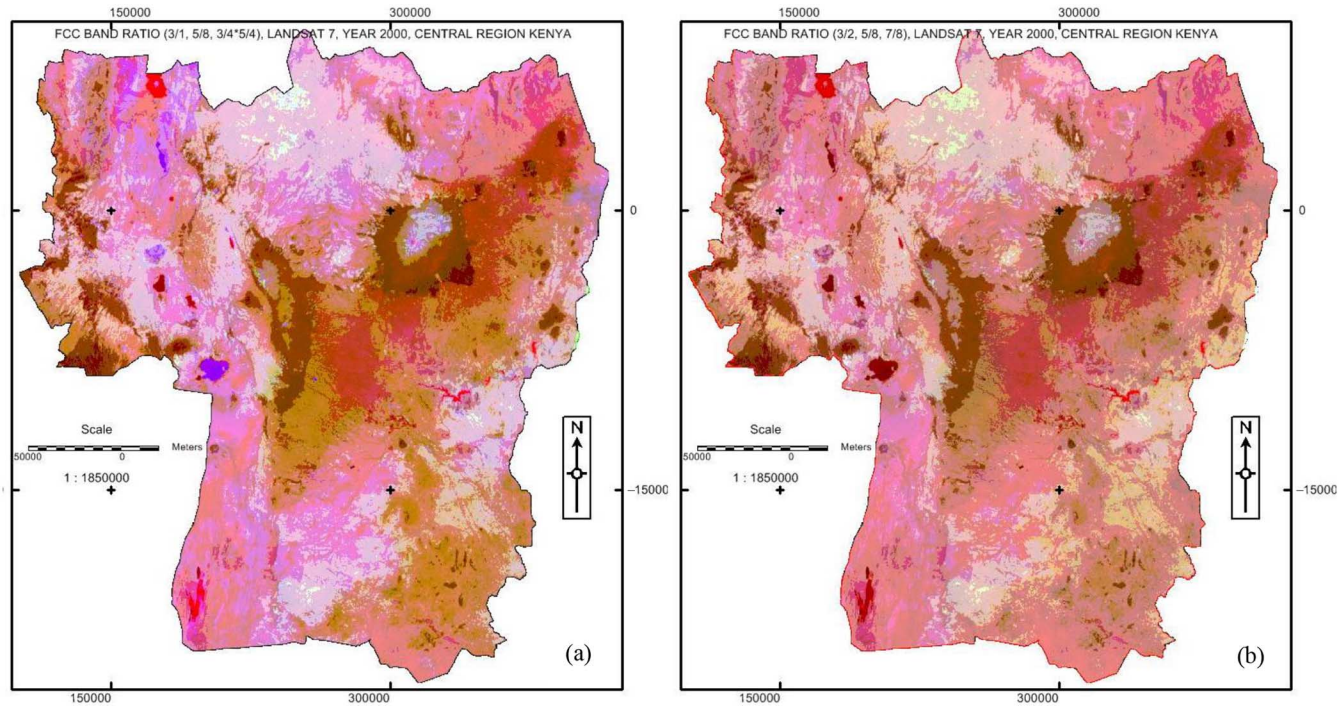

Fig. 8. FCC Band ratios. (a) (3/1, 5/8, $3/4 * 5/4$). (b) (3/2, 5/8, 7/8).

[Fig. 6(c)]. Comparison of Fig. 6(b) and (c) to IHS of FCC 5,7,3 Fig. 6(a) revealed more enhanced visualization of lineaments in Fig. 6(b) and (c).

Band ratio combinations involving bands from different spectral regions were found to have good contrast and thus the following band ratios were possible: 7/3, 7/4, 5/3, 5/1, and 5/4. Additional band ratios involving bands on the same spectral were: 3/1, 3/2, and 7/5, where ratio 3/2 provided important information on water turbidity, while multiplicative ratio $3/4 * 5/4$ was borrowed from [21] and thus their incorporation into FCCs. Since pan-band 8 of Landsat 7 occupies the wavelength of bands 2, 3, and 4, then ratios involving the mid-infrared region: 5/8 and 7/8 were tested. The following FCC were found to emphasize geological features: (5/1, 5/3, 7/4), (3/2, $3/4 * 5/4$, 7/3), (3/2, 5/4, 7/3), (5/1, $3/4 * 5/4$, 7/5), (3/2, 5/1, 7/3), and (3/2, 5/1, 7/4) of Fig. 7(a)–(f), respectively. FCCs involving band 8 are in Fig. 8(a) and (b): (3/2, 5/8, 7/8) and (3/1, 5/8, $3/4 * 5/4$).

C. Geology/Soils Mapping

Geology and soils mapping was achieved by performing knowledge-based classification guided by thresholding of band ratio thresholds as in Table II using band ratios only. The choice of band ratios used in the classification was guided by enhanced contrast in the FCCs Fig. 7(a)–(f) emphasized geological features and texture information in the individual band ratio. Fig. 7(e) and (f) shows the sharpest contrast, thus presenting band ratios 3/2, 5/1, 7/3, and 7/4 as the most suitable for the classification. However, band ratio 7/4 was not used in the classification, since band ratio 7/3 captures the properties from band 7, and band ratio 5/4 had clay minerals more emphasized than in ratio 7/4. Therefore, band ratios (5/1, 5/4, 7/3, and 3/2) were used as input for knowledge-based classification.

TABLE II
KNOWLEDGE-BASED CLASSIFICATION CLASS BOUNDARIES THRESHOLD USING BAND RATIOS TO MAP GEOLOGY

	3/2	5/1	7/3	5/4
Pyroclastic unconsolidated	0.5–1.2	0.5–1.85	>1.150	
Basic metarmophic	1.050–1.35	1.400–1.85	0.675–1.050	
Basic igneous	1.050–1.45	1.050–1.45	0.5–1.050	
Eolian unconsolidated	0.5–1.35	0.1–1.050	0.5–0.850	
Acidic igneous	0.55–1.2	>1.35	0.900–1.050	
Igneous rocks	0.5–1.2	0.5–1.35	0.85–1.150	
Intermediate Igneous	>1.000	>1.800	>1.050	
Fluvial deposits	>1.35	>1.45	0.600–1.2	
Acidic metamorphic	>1.200	1.050–1.800	>1.050	
Shallow water	<0.85	<0.1	<0.5	<0.1
Deep water	0.85–1.4	<0.1		<0.1
Salt bearing rocks	0.85–1.4	0.1–0.65	<0.75	
Water clay deposits		<0.1		<=1

Threshold values in Table II were determined by running advanced RGB clustering (in Erdas Imagine) of FCC (3/2, 5/1, 7/3) with 32 number of classes. The clustering results class boundary values were examined for each band ratio and the classes were refined further by setting threshold class boundaries that combined classes overlapping in all band ratios. Knowledge-based classification was run in Erdas Imagine software, where the classes were set and the class rules specified as in Table II for each of the attribute raster band ratios in the knowledge engineer. The Landsat image for the study area was then input together with the saved knowledge base file to run the classification and the result is shown in Fig. 9(a).

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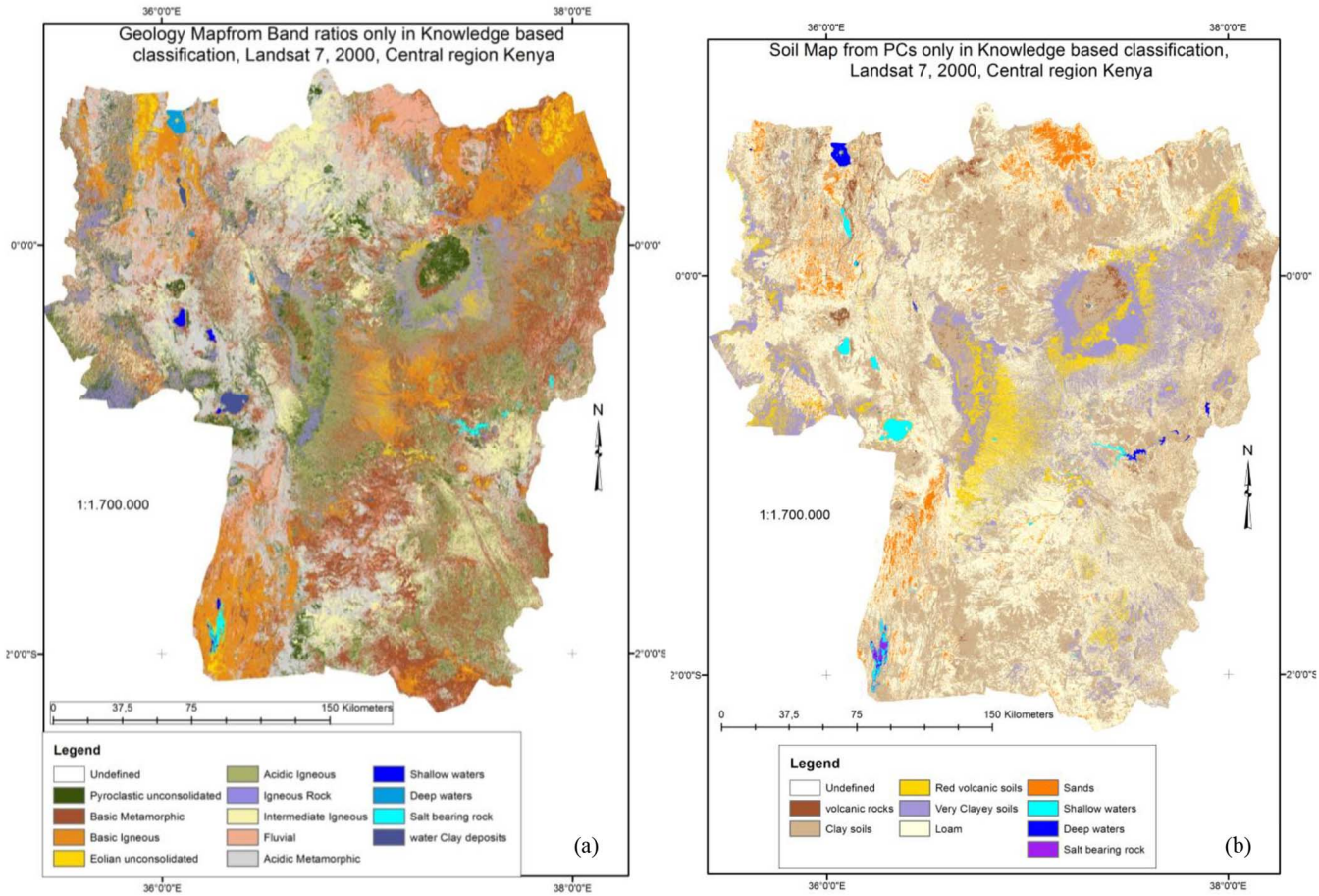

Fig. 9. (a) Geology maps derived from band ratios in knowledge-based classification. (b) Soil map derived from PCs 1, 2, 5 and IC1 in knowledge-based classification using Landsat imagery, year 2000.

TABLE III
KNOWLEDGE-BASED CLASSIFICATION CLASS BOUNDARIES USING PCS TO MAP GEOLOGY

Element		PC2	PC5	PC1	IC1
	Histogram range	–106.162 to +184.04	–57.57 to 57.97	0–559.738	–1.33 to 255.064
Volcanic rocks (agglomerates)		30–80	10–15		
Clay soils		30–80	0–10		
Red volcanic soils		30–80	Zero to –12		
Very clayey soils (Tuff)		<30	2–10		
Loam (volcanic ashes)		<30	2–(–8)		
Sands (sedimentary deposits)		<0	<–9		0–4
Shallow water		30–5	10–15	<85	1–8
Deep water			<5		3–9.5
Salt bearing rocks		>80			>1

A comparison was made by running another classification, guided by abundance and ease of geological features in PCs (2, 3, 4, and 5) with PCs' class boundaries set as in Table III. This was guided by PC factor loading analysis and PCs FCC emphasizing most geologic features with IC1 replacing PC1, which had positive correlation for all bands. Class boundaries threshold values were set after carefully selecting training areas and checking their upper and lower boundary values. PC2 and PC4 had the most geological features, PC3 and PC5 had a lot of water information, PC3 provided soil moisture information, while IC1 was found better at discriminating water types together with PC1. The resulting map from this classification is shown in Fig. 9(b).

TABLE IV
LINEAMENT EXTRACTION THRESHOLD

Sobel output edges	Band 8	Band ratio (5/1)
	(0–319)	(0–1.96)
Lineaments	>30	>0.300

D. Lineament Extraction

The basis for lineament extraction was band ratio with enhanced texture property, in which the band ratio 5/1 was found suited, and increased chances of more edge features where band 8 with finer resolution was most suited. Edges were extracted by application of nondirectional edge-detector

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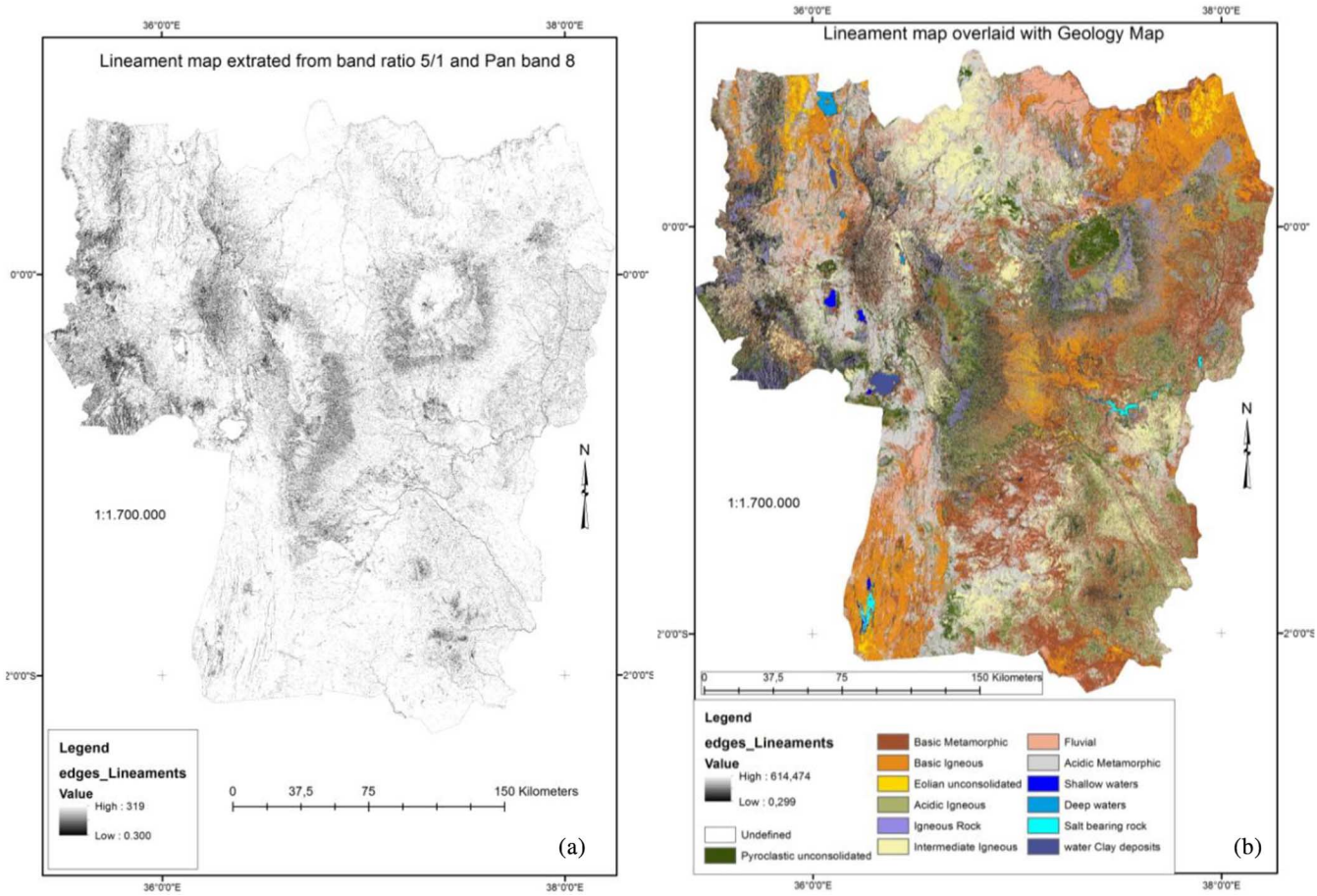

Fig. 10. (a) Lineament map extracted from band ratio 5/1 and pan-sharpened band 8 edges. (b) Geology map overlaid with lineament.

sobel operator to both pan-band 8 and band ratio 5/1. For slight enhancement of the edges, a multiplicative factor of 3 was used in the sobel operator. The edge files obtained from the application of the filter were then the input variables in the knowledge-based classification, where threshold values were set as in Table IV. By applying an edge-directional filter, homogeneous areas are smoothed out while edges and linear features were more enhanced. Thresholding ensured that only major linear features are selected in the classification and the result is shown in Fig. 10(a). More refinement of the lineaments was done to join point features to line features and the results overlaid with the classification geology map [Fig. 10(b)].

Another method that was found to emphasize lineament features was extracting edges from bands 5 and 8 using sobel edge detector and combining them in RGB combination where slope was the third band (Fig. 11).

IV. Results and Discussion

Sharp contrast was observed in FCC (5/1, 5/3, 7/4), (3/2, 5/1, 7/3), and (3/2, 5/1, 7/4), i.e., Fig. 7(a), (e), (f), while texture information was much more pronounced in FCCs (3/2, 5/1, 7/3) and (3/2, 5/1, 7/4). Water types and turbidity types were more emphasized in FCCs: (5/1, 5/3, 7/4), (3/2, $3/4 * 5/4$, 7/3), (3/2, 5/4, 7/3), and (3/2, 5/1, 7/4), while FCC (5/1, $3/4 * 5/4$, 7/5) did not map out any fresh or shallow water bodies. FCCs involving pan-sharpened band 8 and mid-infrared bands presented in Fig. 8 had bare volcanic rocks and bare soils well highlighted against moist vegetated regions. FCC (3/1, 5/8, $3/4 * 5/4$) differentiated wet areas from water bodies better than FCC (3/2, 5/8, 7/8) (Fig. 8).

It was observed that geology contrast was increased when a higher band was divided by a lower band, as Drury [33] defined band rationing. Thus, while it was possible to have ratios involving a lower band divided by a higher band, e.g., 5/7, 3/4, and 4/5, these combinations resulted in emphasized vegetated regions since the study area has both highlands and semiarid characteristics. Hence, ratios involving lower band versus high band were not used in this study.

It was also noted that band ratios involving bands 4 and 2 as the numerator resulted in emphasized vegetation features while their use as denominator resulted in emphasized clay minerals and water turbidity information, respectively. Thus, band ratios 4/3, 4/5, 2/3, or 2/1 were eliminated. Given that band ratio FCC requires atleast three different bands, and that Landsat has possible six bands, then it was possible to obtain 20 FCC band ratios by combinations and permutations algebra [34]. However, only the FCCs presented in Fig. 7 had good contrast to qualify in the classification criteria.

It can be noted from the resulting geology classification map [Fig. 9(a)] that although the individual FCC combinations presented in Fig. 7 had good contrast, each combination

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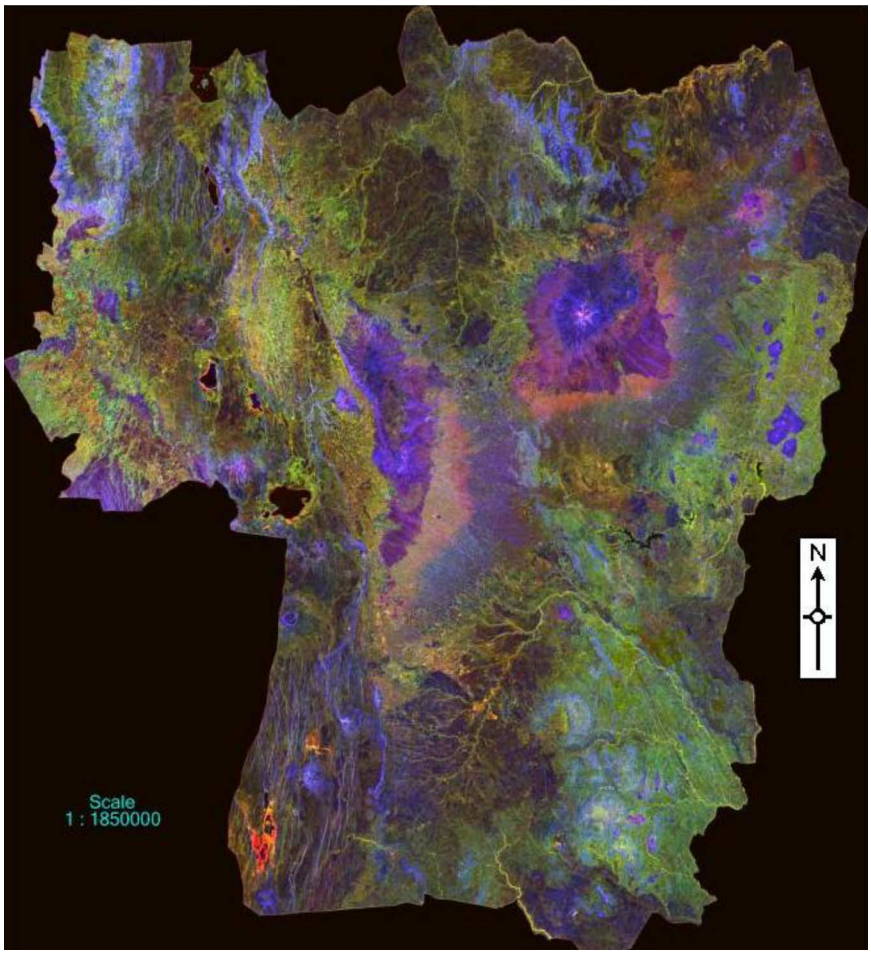

Fig. 11. Lineament map extracted from band 5, band 8, and slope.

had specific features emphasized more than others and thus, knowledge-based classification result captured the strength of each band ratio. The combination of the bands used in the classification captured all Landsat 7 bands except band 6 and pan-band 8, and the numerators were bands 3, 5, and 7 [i.e., Fig. 3(a)], as they contained the most geological information in the factor loading. However, the FCC involving band 8 compares to PC classification [Fig. 9(b)] with wetness being the key denominator.

The FCC composites (3/1, 5/8, $3/4 * 5/4$) and (3/2, 5/8, 7/8) in Fig. 8 had higher spatial resolution but lower contrast compared to composites (3/2, 5/1, 7/3), (3/2, 5/4, 7/3), or (3/2, 5/1, 7/4). This could be explained by the fact that band 8 occupies the spectral region of bands 4, 3, and 2 in Landsat 7 and thus, the FCCs contain redundancy in the band ratio denominator. The combination of band ratios 3/2 and 5/1 emphasized all classes of igneous rocks, where ratio 3/2 was instrumental in emphasizing the iron oxides (ferro-magnesian minerals) present in the volcanic rocks [Fig. 7(e) and (f)]. Acidic metamorphic (quartzite, gneiss, migmatite) and pyroclastic unconsolidated rocks were emphasized by the combination of band ratios 7/3 and 3/2 [Fig. 7(b), (c), and (e)], whereas the combination of band ratios 7/3 and 5/1 emphasized eolian unconsolidated and basic metamorphic rocks [Fig. 7(a), (e), and (f)]. Band ratio 5/8 achieved some similar effect as ratio 7/3 in emphasizing acidic metamorphic rocks and intermediate igneous rocks (Figs. 7 and 6(b) and (c)], and ratio 5/1 differentiated the two classes [Fig. 7(e)]. The use of multiplicative band ratio $3/4 * 5/4$ resulted in the loss of sharp distinction between basic igneous (basalts) and basic metamorphic (gneiss) rocks, while clay deposits in water were not mapped [Fig. 7(b), (d)]. Water clay deposits were emphasized by ratios 5/4 and 3/2 [Fig. 7(b), (c)], while shallow water beds were emphasized by ratios 5/1 and 3/2.

Results obtained from band ratio classification [Fig. 9(a)] had more classes compared to results obtained from PC classification [Fig. 9(b)]. This may be explained by the fact that PC works by reducing the number of bands in the original information [18], while band rationing uses the original bands to emphasize the mineral element present in the rock. It was, therefore, more difficult to differentiate certain elements in the PC classification that were well differentiated in the band ratio classification. Band ratio classification matched existing geology map and filled the missing gaps in the vector map (Fig. 1).

Fig. 10(a) shows the lineament map extracted after filtering applications of band ratio 5/1 and pansharpened band 8. The result highlighted features of relief and drainage as well as possible fault lines. However, there was a challenge visualizing the lineaments by incorporating them into the geology map and instead, an overlay of the geology map with lineament

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was performed [Fig. 10(b)]. Also, most information was in point form rather than lines especially when viewed at large scale. Kavak [18] described a similar lineament extraction procedure using LINE module of PCI Geomatica. However, the researcher recommended the definition of orientation direction of most lineaments making it difficult in situation with high relief features.

The lineament map overlaid with geology map in Fig. 10(b) was compared with Fig. 6(b), where lineament features are emphasized by combining IC1, PC2, and saturation band of IHS FCC 573. This idea was borrowed from Mondini *et al.* [35] who identified landslide areas using RGB combination comprising change in NDVI, IC1, and PC1. In this case, components of both PCA and ICA containing most geological information were used together with saturation band of the FCC containing most geological information. The results [Fig. 6(b) and (c)] had lineaments more emphasized than PC combinations [Fig. 2(a) and (b)] or fused edges with FCC 573 [Fig. 5(b)]. It was noted that Fig. 6(c) involving FCC (IC1, PC5, saturation band of IHS 573) had the best discrimination of geological features closely matching the classification map from band ratios and better visualization of the lineaments. Volcanic extrusions appeared in light green, igneous rocks appeared in blue, sedimentary rocks (eolian unconsolidated, organic) appeared in red to hot pink colors, fluvial deposits appeared in purple–magenta colors, and water appeared white to light pink color with increasing turbidity.

Fig. 11 shows an alternative lineament map obtained by RGB combination of edges from band 5, 8, and a slope map of the study area. The map emphasized lineaments especially along the Rift valley and high relief features. This was due to the contribution of the slope element; otherwise, the edges are not as sharp as in Fig. 10(a).

V. Conclusion and Recommendation

The choice of band ratios 3/2, 5/1, 7/3, and 5/4 utilized all the possible Landsat 7 bands thereby, enabling the strength of each band to emphasize mineral elements comprising the geological features. Their combinations had more contrast compared to the PC combinations, a reason which may have contributed to the resulting geology map having more classes than the one obtained from the PC classification map. This may support the use of band ratios in applications requiring more precise mapping and sharp distinction of elements especially with the availability of hyperspectral data, where an element can be studied in several narrower spectral bands. In general, the band ratio FCC contrast improved with lack band redundancy in both numerator and denominator, while use of band 8 in band ratios merged information from the bands where they overlap (i.e., 2–4). However, the utilization of band 8 may form a basis for soil wetness mapping.

The lineaments obtained coincided well with the existing drainage features and when overlaid with the geological map, rock types were emphasized along the boundaries. Lineament features were more pronounced in the FCC combination involving IC1, PC5, and saturation band of IHS FCC (5, 7, 3) compared to PC combinations or fused edges with FCC (5,7,3). Complex folding and high density of lineament features along the rift valley and high relief features, respectively, and lineament orientation from enhanced texture information were well visualized. This will be investigated for further landslide factor analysis especially relating to changes as a result of landslide deposition or exposed intrusive rocks.

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References

[1] I. (AGS) Auracle Geospatial Science. (2011, Nov.). *Remote Sensing for Geological Mapping and Mineral Exploration – Spectral Imaging – (Part 3 of 4) Remote Sensing and Geospatial Intelligence News* [Online]. Available: http://auracle.ca/news/?p=155, accessed on Jul. 12, 2013.

[2] R. P. Gupta, "Geological applications," in *Remote Sensing Geology*, 2nd ed. Berlin, Germany: Springer-Verlag, 2003, pp. 429–583.

[3] M. M. Abdeen and A. A. Abdelghaffar, "Mapping Neoproterozoic structures along the central Allaqiheiani suture, Southeastern Eqypt, using remote sensing and field data," presented at the 29th Asian Conf. Remote Sens., Colombo, Sri Lanka, 2008, vol. 3.

[4] P. C. Kariuki, T. Woldai, and F. Van der Meer, "The role of remote sensing in mapping swelling soils," *Asian J. Geoinformat.*, vol. 5, no. 1, pp. 43–54, 2004.

[5] T. C. Davies, "Landslide research in Kenya," *J. Afr. Earth Sci.*, vol. 23, no. 4, pp. 541–545, Nov. 1996.

[6] T. C. Davies and I. O. Nyambok, "The Murang'a landslide, Kenya," *Environ. Geol.*, vol. 21, pp. 19–21, 1993.

[7] M. W. Ngecu and M. E. Mathu, "The El Nino triggered landslides and their socio- economic impacts on Kenya," *Episodes*, vol. 22, no. 4, pp. 284–289, Dec. 1999.

[8] W. M. Ngecu, C. M. Nyamai, and G. Erima, "The extent and significance of mass-movements in Eastern Africa: Case studies of some major landslides in Uganda and Kenya," *Environ. Geol.*, vol. 46, no. 8, pp. 1123–1133, Jul. 2004.

[9] L. Q. Hung, O. Batelaan, and F. De Smedt, "Linearment extraction and analysis, comparison of Landsat ETM+ and ASTER imagery. Case study: Suoimuoi tropical karst catchment, Vietnam," in *Proc. SPIE*, 2005, vol. 5983.

[10] A. R. Newton and T. P. Boyle, "Discriminating rock and surface types with multispectral satellite data in the Richtersveld, NW Cape Province, South Africa," *Int. J. Remote Sens.*, vol. 14, no. 5, pp. 943–959, Mar. 1993.

[11] M. M. Abdeen and S. M. Hassan, "Utilisation of spectral signature and principal component analysis, of TERRA TERRA ASTER images for exploring new sites of building sand and gravels, Northwest Gulf of Suez, Egypt," presented at the 30th Asian Conf. Remote Sens., 2009.

[12] I. D. Novak and N. Soulakellis, "Identifying geomorphic features using Landsat-5/TM data processing techniques on Lesvos, Greece," *Geomorphology*, vol. 34, no. 1–2, pp. 101–109, Aug. 2000.

[13] E. A. Ali, S. O. El Khidir, A. A. Babikir, and E. M. Abdelrahnam, "Landsat ETM+7 digital image processing techniques for lithological and structural lineament enhancement: Case study around Abidiya Area, Sudan," *Open Remote Sens. J.*, vol. 5, no. 1, pp. 83–89, Aug. 2012.

[14] F. F. Sabins, *Remote Sensing: Principles and Applications*, 3rd ed. San Francisco, CA, USA: Freeman, 1997.

[15] N. A. Al Muntshry, "Evaluating the effectiveness of Multispectral Remote Sensing data for Lithological Mapping in arid regions: A quantitative approach with examples from the Makkah neoproterozoic region, Saudi Arabia," M.Sc. Thesis, Missouri Univ. Sci. Technol., Rolla, MO, USA, 2011.

[16] A. P. Crósta and J. M. Moore, "Geological mapping using Landsat Thematic Mapper imagery in Almeria Province, south-east Spain," *Int. J. Remote Sens.*, vol. 10, no. 3, pp. 505–514, Mar. 1989.

[17] M. Wahid and R. E. Ahmed, "Identifying geomporphic features between Ras Gemsha and Safaga, Red Sea Coast, Egypt, using remote sensing techniques," *Mar. Geol.*, vol. 17, no. 1, p. 23, 2006.

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[18] K. S. Kavak, "Determination of palaeotectonic and neotectonic features around the Menderes Massif and the Gediz Graben (western Turkey) using Landsat TM image," *Int. J. Remote Sens.*, vol. 26, no. 1, pp. 59–78, Jan. 2005.
[19] J. A. Richards and X. Jia, "Interpretation of hyperspectral image data," in *Remote Sensing Digital Image Analysis*, 4th ed. Berlin, Germany: Springer-Verlag, 2006, pp. 359–388.
[20] B. Mia and Y. Fujimitsu, "Mapping hydrothermal altered mineral deposits using Landsat 7 ETM+ image in and around Kuju volcano, Kyushu, Japan," *J. Earth Syst. Sci.*, vol. 121, no. 4, pp. 1049–1057, Aug. 2012.
[21] M. Sultan, R. E. Arvidson, and N. C. Sturchio, "Mapping of serpentinites in the Eastern Desert of Egypt by using Landsat thematic mapper data," *Geology*, vol. 14, no. 12, p. 995, 1986.
[22] F. F. Sabins, "Remote sensing for mineral exploration," *Ore Geol. Rev.*, vol. 14, no. 3, pp. 157–183, Sep. 1999.
[23] J. B. Campbell, *Introduction to Remote Sensing*, 2nd ed. New York, NY, USA: Guildford Press, 1996.
[24] E. R. Crippen, "Selection of Landsat TM band and band-ratio combinations to maximize Lithologic information in color composite displays," in *Proc. 7th Thematic Conf. Remote Sens. Explor. Geol.*, Calgary, Alberta, 1989, pp. 917–921.
[25] N. H. Kenea, "Improved geological mapping using Landsat TM data, Southern Red Sea Hills, Sudan: PC and IHS decorrelation stretching," *Int. J. Remote Sens.*, vol. 18, no. 6, pp. 1233–1244, Apr. 1997.
[26] J. R. Harris *et al.*, "Remote sensing as a geological mapping tool in the Arctic: Preliminary results from Baffin Island, Nunavut," *Curr. Res.*, vol. 2001–E12, p. 13, 2001.
[27] S. A. Rawashdeh, B. Saleh, and M. Hamzah, "The use of remote sensing technology in geological investigation and mineral detection in El Azraq-Jordan," *Eur. J. Geogr.*, no. 358, Oct. 2006, doi: 10.4000/cybergeo.2856.
[28] M. H. T. Qari, A. A. Madani, M. I. M. Matsah, and Z. Hamimi, "Utilization of Aster and Landsat data in geologic mapping of basement rocks of Arafat area, Saudi Arabia," *Arab. J. Sci. Eng.*, vol. 33, no. 1C, pp. 99–117, Jun. 2008.
[29] A. Kocal, H. S. Duzgun, and C. Karpuz, "Discontinuity mapping with automatic lineament extraction from high resolution satellite imagery," in *Proc. 20th ISPRS Congr.*, Istanbul, Turkey, 2004 [Online]. Available: http://www.isprs.org/istanbul2004/comm7/papers/205.pdf
[30] M. W. Mwaniki, T. G. Ngigi, and E. H. Waithaka, "Rainfall induced landslide probability mapping for central province," in *Proc. 4th Int. Summer Sch. Conf.*, Kenya, 2011, vol. 1, pp. 203–213.
[31] M. W. Ngecu and D. W. Ichang'i, "The environmental impact of landslides on the polulation living on the eastern footslopes of the Aberdare ranges in Kenya: A case study of Maringa Village landslide," *Environ. Geol.*, vol. 38, no. 3, pp. 259–264, Sep. 1998.
[32] M. Ehlers *et al.*, "Spectral characteristics preserving image fusion based on Fourier domain filtering," in *Proc. SPIE Remote Sens. Environ. Monit. GIS Appl. Geol. IV 1*, Canary Islands, Spain, 2004, vol. 5574, pp. 1–13.
[33] S. A. Drury, *Image Interpretation in Geology*, 2nd ed. London, U.K.: Chapman and Hall, 1993.
[34] M. Bóna, *Combinatorics of Permutations*. Boca Raton, FL, USA: Chapman and Hall/CRC, 2004.
[35] A. C. Mondini, K.-T. Chang, and H.-Y. Yin, "Combining multiple change detection indices for mapping landslides triggered by typhoons," *Geomorphology*, vol. 134, no. 3–4, pp. 440–451, Nov. 2011.

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